AI EXPOSURE IN THE PHILADELPHIA METRO LABOR MARKET

Industry and occupational patterns

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Introduction

This report looks at observed AI exposure by occupation to answer 3 main questions about the Philadelphia metro labor market: Which industries and occupations are already affected by AI? Who holds these jobs? Furthermore, is any of this showing up in employment numbers yet? Observed AI exposure shows where AI is already integrated in daily tasks. High exposure means AI already covers a large share of the tasks it could do in that job, while low exposure means AI covers little of what it could do there.

This analysis uses the **observed AI exposure index** from the Anthropic Economic Index (March 2026) and is inspired by the research from the Yale Budget Lab. For each of the 756 detailed occupations, the percentage shows how many of the tasks AI could do are actually being done with AI. Another measure looks at usage density, or how much total AI activity comes from workers in each occupation. The data comes from The Budget Lab at Yale (June 2026) and uses the same Anthropic dataset that compares each occupation to the workforce average.

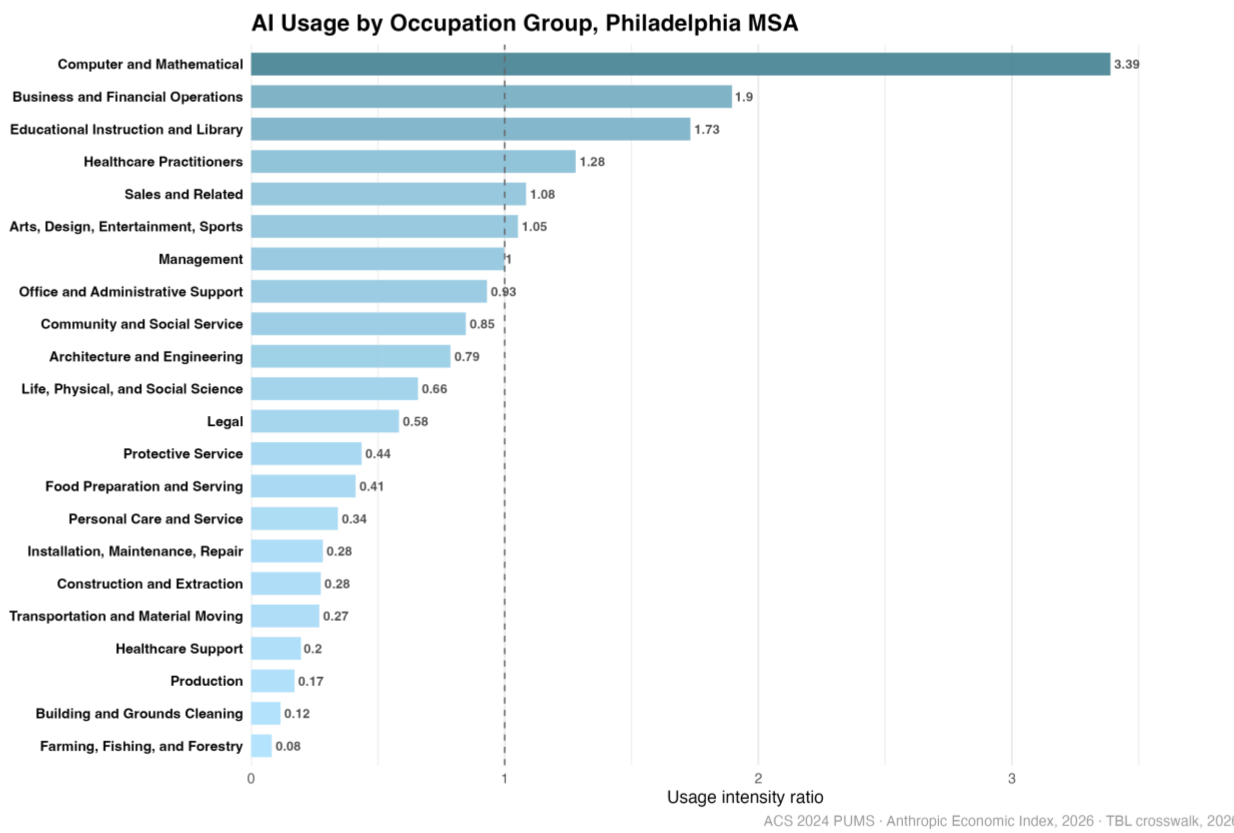

Figure 1. AI usage by occupation group, Philadelphia MSA

Figure 1 ranks the 22 major occupation groups by usage density. Computer and mathematical workers stand far apart, at 3.39 times the workforce average. Business and financial operations come second at 1.90, followed by educational instruction (1.73) and healthcare practitioners (1.28). Education and health care also employ more people than any other part of the region's workforce. Teachers and clinicians generate more AI activity than the average across the workforce. Farming registers 0.08 times the average; building and grounds cleaning, 0.12; production, 0.17. Office and administrative support lands near the middle at 0.93.

Each of the region's 3.26 million workers is matched to an exposure level based on their detailed occupation in the 2024 ACS 1-year PUMS. This means the local geography reflects the mix of jobs people have, and PUMS shows where residents live, not where residents work. To check the results, industry rankings were also calculated using three published studies (Felten, Raj & Seamans 2021; Eloundou et al., 2023; Tomlinson et al., 2025).

What the Philadelphia data shows

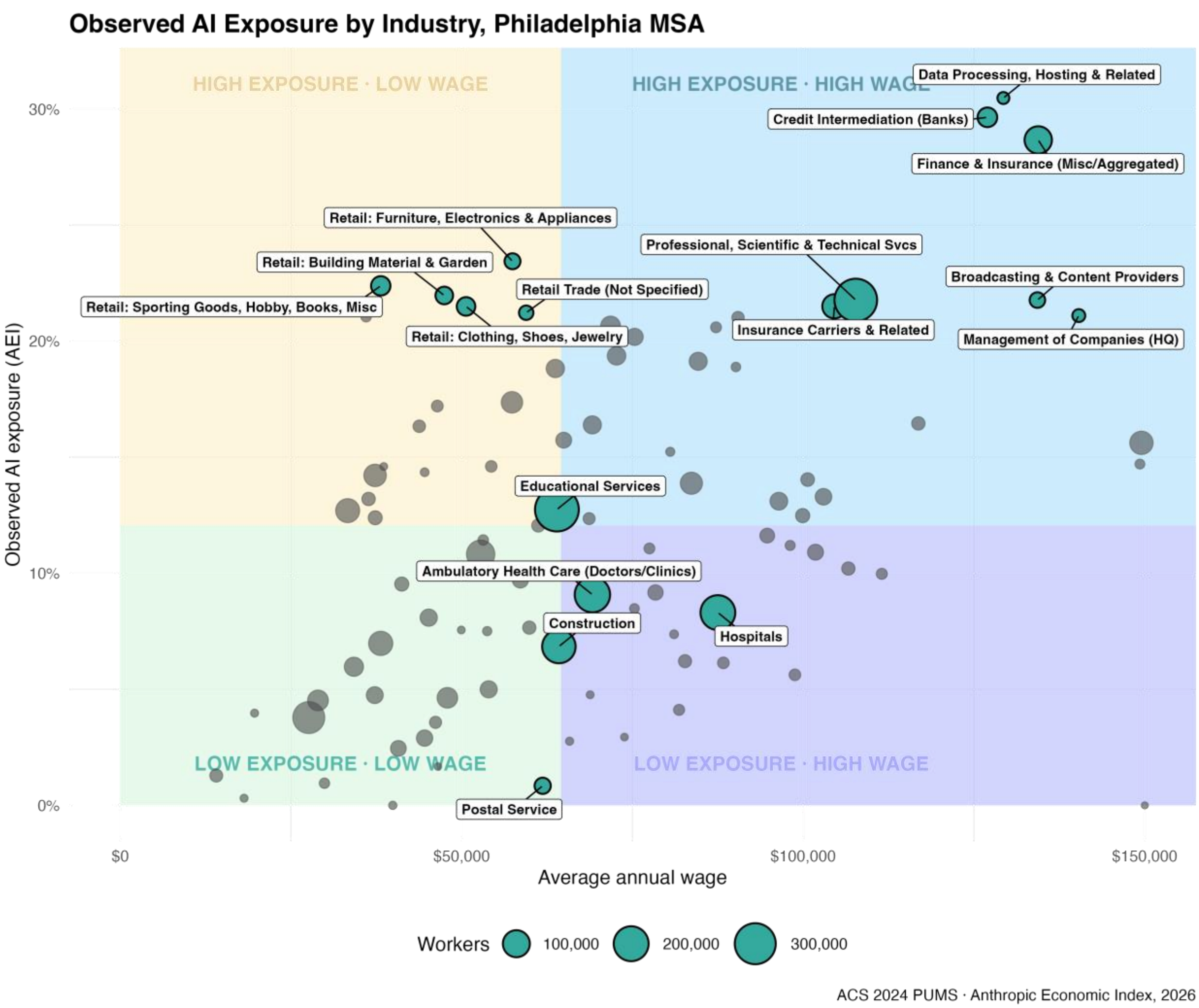

Figure 2. Observed AI exposure by industry, Philadelphia MSA

Finance has the highest AI exposure, but retail stands out. Industry exposure ranges from almost zero (Postal Service) up to about 31% (Data Processing). Credit Intermediation (Banks) and Finance & Insurance are at the top. Less expected is that retail industries such as clothing, furniture, electronics, building materials, and sporting goods have exposure rates between 21% and 24%, much higher than the metro median of about 12%. This happens because sales jobs have a limited set of tasks that AI can perform, and those tasks, such as clarifying product questions, promoting, writing, advertising, and conducting inventory checks, are where AI can be used most.

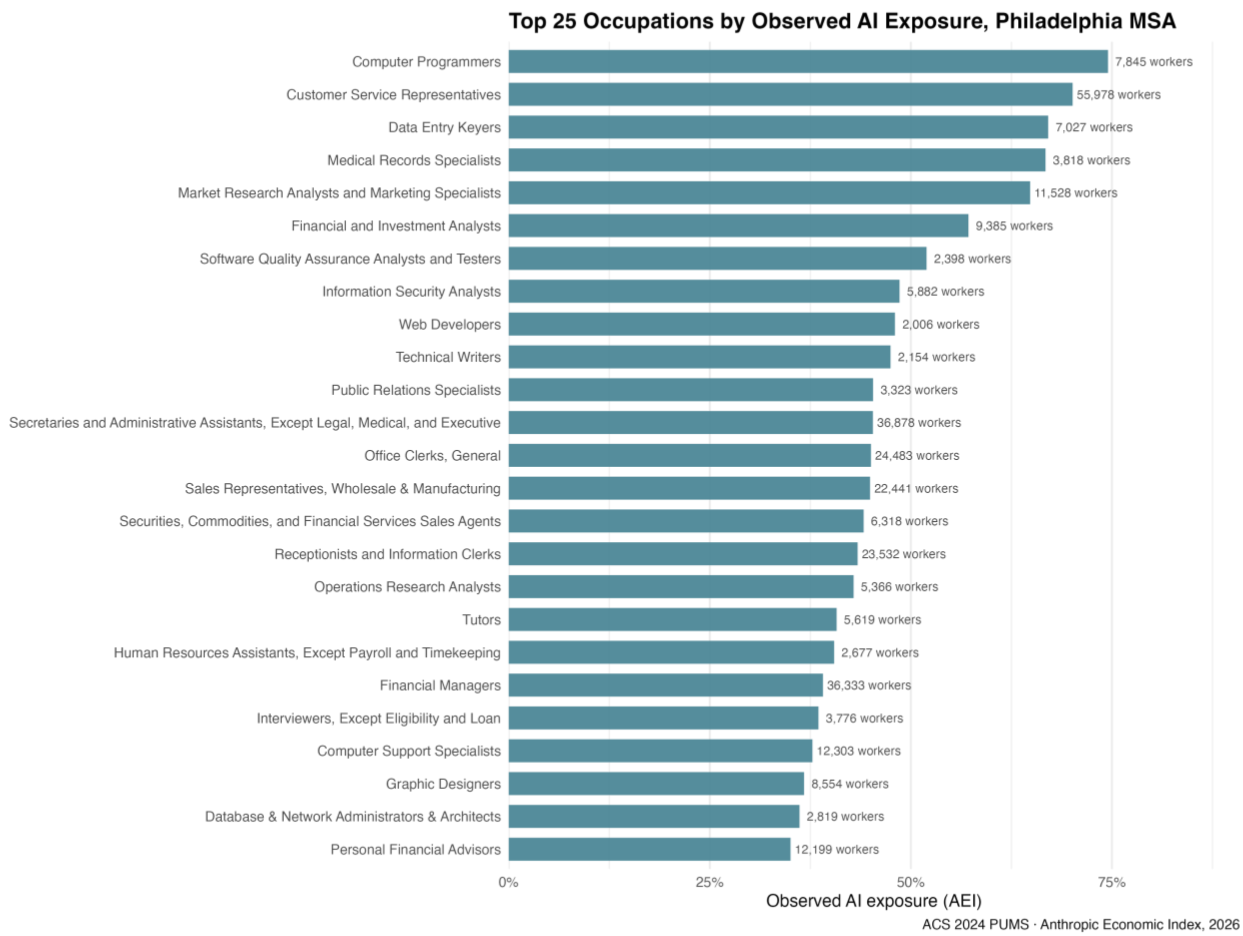

Figure 3. Top 25 occupations by observed AI exposure, Philadelphia MSA

Sales Representatives (Wholesale & Manufacturing), with 22,000 workers, rank 14th among the top 25 occupations (Figure 3), at about 45% exposure. Customer Service Representatives (56,000 workers, about 70%), Secretaries and Administrative Assistants (37,000, about 44%), and Office Clerks (24,000, about 43%) follow. At the top, Computer Programmers have about 74% exposure. The region's deepest AI use is in office, sales, and retail jobs, although tech jobs are only a small part.

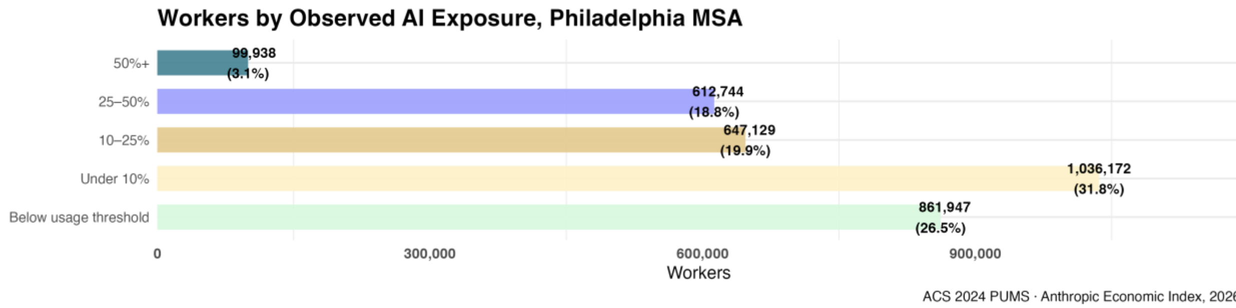

Figure 4. Workers by observed AI exposure, Philadelphia MSA

More than one in five workers already hold jobs with high AI exposure. 27% of workers are in jobs below the usage-reporting threshold, 32% of workers are under 10% AI exposure, 20% are between 10% and 25% AI exposure, and 22% (roughly 713,000 people) are in jobs where more than a quarter of possible daily tasks already use AI. Of these, 100,000, or 3.1% of all workers, are in jobs where over half of tasks use AI.

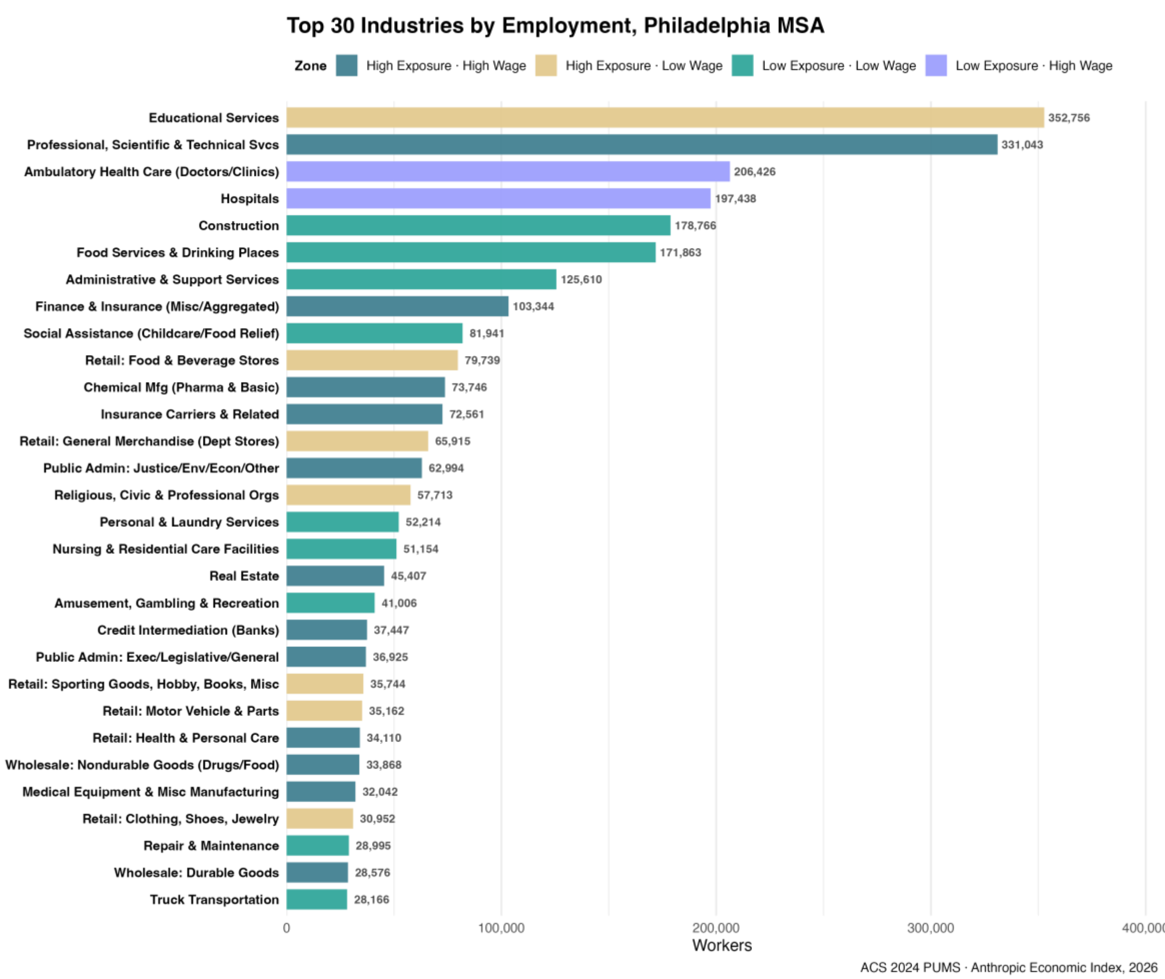

Figure 5. Top 30 industries by employment, Philadelphia MSA

Most high-exposure industries also sit on the high-wage side of Figure 2. Retail concentrates the high-exposure, low-wage jobs. These workers are already seeing a lot of AI in their jobs but have lower average wages. Food & Beverage Stores (80,000 workers), General Merchandise (66,000 workers), and other smaller retail groups together employ several hundred thousand workers in this category.

The neighborhood map shows how jobs are spread across the region. On average, exposure is lowest, about 8%, in North and West Philadelphia, and highest, over 15%, along the Main Line, Conshohocken, King of Prussia, and central Bucks County. The city's lower exposure reflects the jobs its residents hold, mostly healthcare support, food service, personal care, and transportation, occupations with almost no observed AI exposure. When mapped by residence, AI exposure follows the region's existing patterns of jobs and income (Figures 6 and 7).

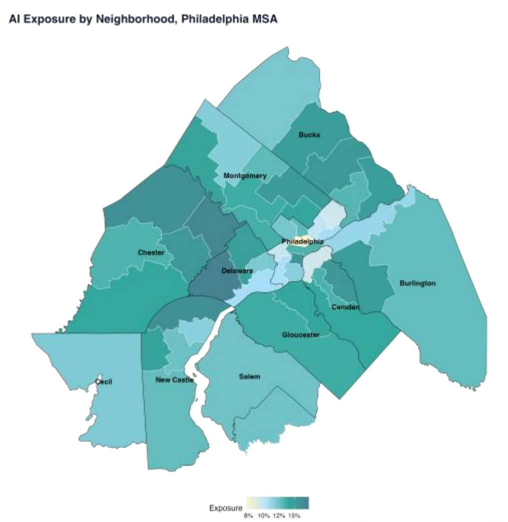

Figure 6. AI exposure by neighborhood, Philadelphia MSA

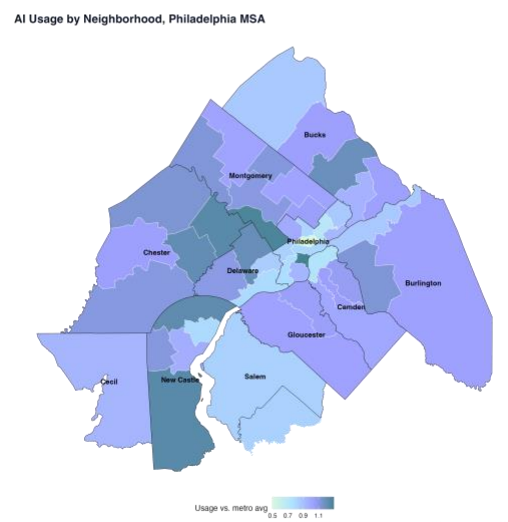

Figure 7. AI usage by neighborhood, Philadelphia MSA

Women have higher AI exposure than men in the Philadelphia MSA, mainly because of the types of jobs they hold. Employment-weighted exposure is 14.7% for women and 11.8% for men on average. Women's exposure ranges from about 15% to 32% across most industries, since many women work in administrative and business roles. For men, exposure ranges from under 5% (in construction, trucking, and repair) to nearly 30% (in banks and data processing). In Management of Companies, Data Processing, and Finance & Insurance, men's exposure exceeds women's.

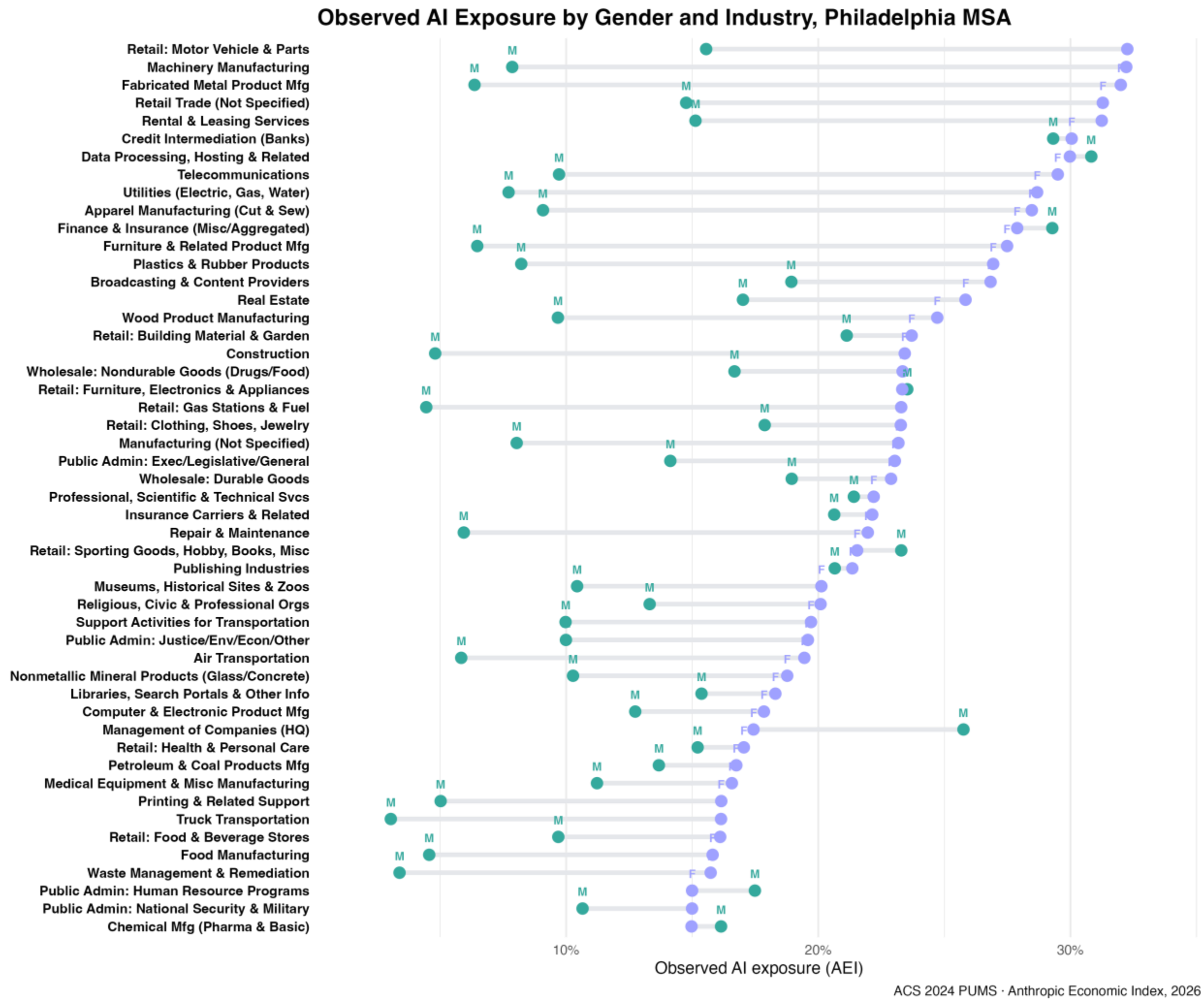

Figure 8. Observed AI exposure by gender and industry, Philadelphia MSA

The specific occupations men concentrate in, such as financial managers and securities agents, carry higher exposure than the occupations women concentrate in. However, it shows a division of jobs instead of anything about the workers. This pattern is similar in other MSAs. The gender gap is smallest where men also have high-exposure jobs (Washington +10%, Boston +12%, Seattle +14%) and largest where men mostly have physical jobs (Dallas +31%, Los Angeles +32%, Miami +38%). Philadelphia, with a 2.9 percentage-point gap, or 24% in relative terms, is in the middle.

Higher exposure does not mean women use AI more. Exposure refers to the related tasks in a job, not the person doing them. Anthropic's survey finds that, within the same job, women use AI in less automated and more step-by-step ways than men (Massenkoff et al., June 2026).

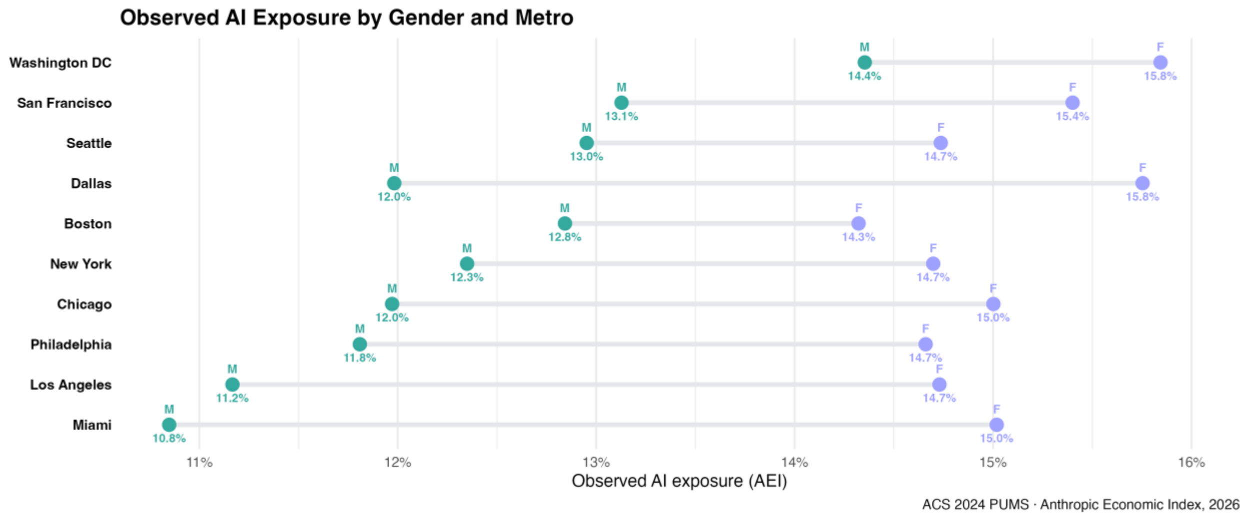

Figure 9. Observed AI exposure by gender and metro

Philadelphia MSA vs Other MSAs

Across ten large metros, employment-weighted AI exposure runs in a narrow band, from 12.8% in Miami and Los Angeles to 15.1% in Washington, DC. Philadelphia's 13.2% ranks next to New York (13.5%) and Chicago (13.4%), a sign of a broad, diversified economy. The tech-heavy metros hold the top of the range, but not by much. In every metro, 21-25% of the workforce holds jobs with more than 25% exposure. That totals 713,000 people in Philadelphia, 2.3 million in New York, and 847,000 in Washington (Figure 10).

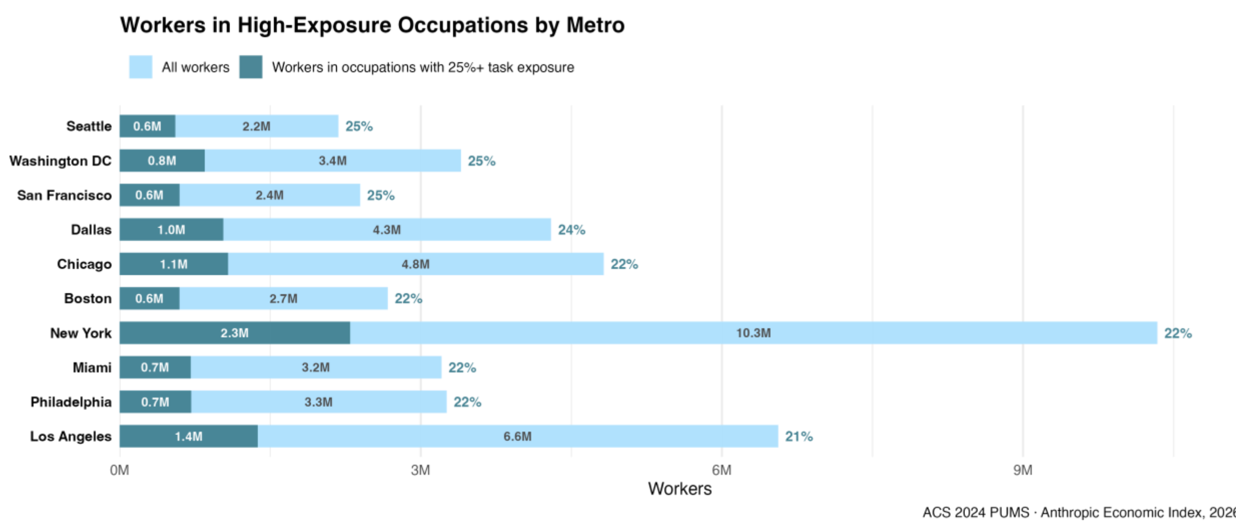

Figure 10 Workers in high-exposure occupations by MSA

Industry rankings change very little across metro areas. Finance & Insurance and Credit Intermediation are always the top two, while retail, wholesale, and professional services make up the next group with only small local changes. Differences in average exposure between metropolitan areas stem from the size of these industries in each place. For example, Insurance Carriers & Related is third in Miami, its highest spot anywhere. Real Estate is ninth in Miami but twelfth or lower in Washington and New York. Retail sectors that are usually third to fifth drop lower in Miami and Dallas. When the ranking chart changes, it is because the mix of job conditions in that industry is different locally.

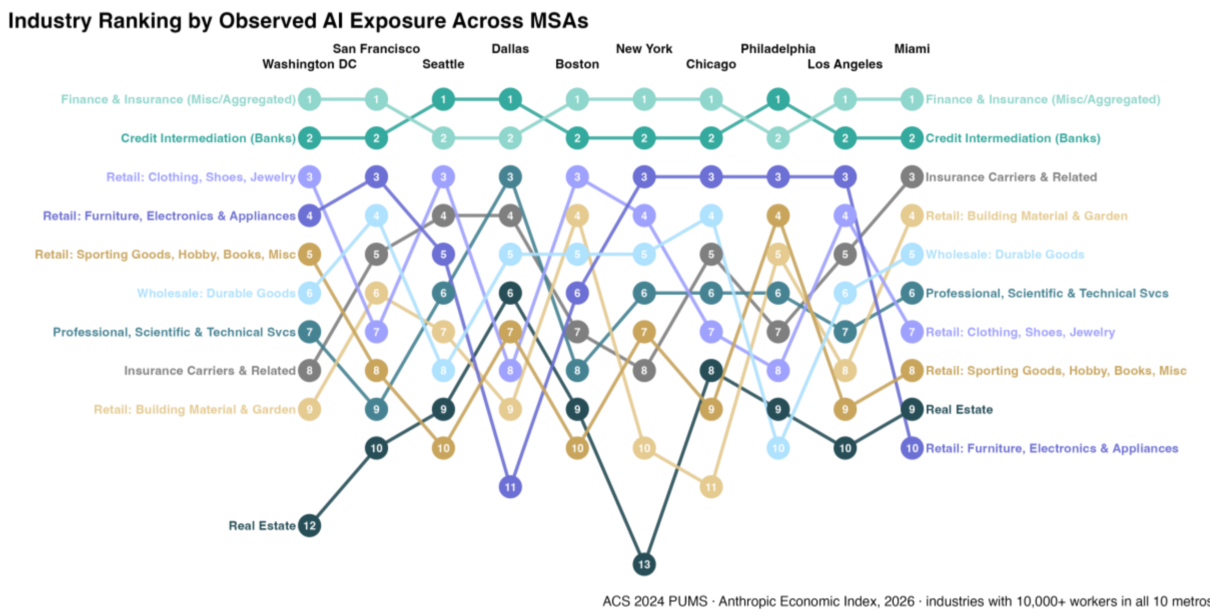

Figure 11. Industry ranking by observed AI exposure across MSAs

At the industry level, the link between exposure and job growth is slightly negative but not statistically significant (Spearman $\rho = -0.21$, $p = 0.07$, across 79 industries). Exposed and unexposed jobs differ for reasons unrelated to AI. Workers in exposed jobs are more likely to have degrees, and these jobs have been less affected by the business cycle in the past (Gimbel, Kendall & Nunn, 2026). Trends like these cannot separate AI's effect from other factors, and averages might hide changes if some exposed jobs grow while others shrink. That is why this report only describes the data.

Through early 2026, there is no significant effect of AI on employment or wages in exposed occupations, and other labor market indicators remain within normal ranges (Gimbel, Kendall & Nunn, 2026; The Budget Lab at Yale, June 2026). Philadelphia's trends fit that national picture. AI use is growing faster than any labor market change. The exposure data represents usage from late 2025. The employment data runs through the end of 2025 and shows little movement since then. Whether the gap between AI adoption and its effects will continue is something the next data releases will help answer.

Limitations

According to Anthropic, AI Exposure is measured nationally at the occupation level. Applying it locally assumes that the same occupation uses AI at the same rate across all metros, which may not hold. However, state-level evidence suggests that occupational composition explains most of the geographic variation in usage (Appel, McCrory & Tamkin, 2025). The measure also has a denominator effect. Occupations where AI can do only a few tasks can show high exposure once those few tasks are covered, so high exposure in retail reflects deep coverage of a narrow task set. The occupation structure comes from 2024 ACS data while the usage data reflects late 2025, a small timing lag.

Data sources

Anthropic Economic Index, March 2026 release.
The Budget Lab at Yale, AI Labor Market Tracker, June 15, 2026 release.
U.S. Census Bureau, American Community Survey 2024 1-year PUMS.
U.S. Census Bureau, TIGER/Line boundaries.
U.S. Bureau of Labor Statistics, Quarterly Census of Employment and Wages, county-level files, 2022 Q1–2025 Q4 (Q4 preliminary)